

SoSe 26 Computer Graphics Solutions

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Sheet 1

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1.1 Assignment 1.1

a) $|\text{Pixel}| = 3840 \times 2160 = 829440$

The whole display is generated in $t = \frac{1}{f} = \frac{1}{60} s$

$\Rightarrow$ Generation of a single pixel can take at most:

$$t_{max} = \frac{1}{3840 \times 2160 \times 60} s \approx 2ns$$

b) Data load of 24 bits per pixel is due to three color channels Red, Green, and Blue, each 8 bit, i.e. 1B. Therefore there are 8 Bytes per pixel and 3840×2160 pixels in total. Since the display is regenerated every $1/60$ seconds, we have:

$$B = 3B \times 3840 \times 2160 \times 60 \frac{1}{s} \\ \approx 1.4GB/s$$

1.2 Assignment 1.2

a) There are three types of light-sensitive receptors,

- S: corresponding to blue
- M: corresponding to green
- L corresponding to red but the perception of color is calculated simply as superposition of three channels but rather as two channels combination of pairs of opposing colors

- b) To focus (or zoom in this context) means to reduce the field of vision, and reducing the field of vision corresponds to a smaller solid angle. The smaller the solid angle is, the less light arrives at the camera lens or human eye through the telescope, which makes the object appear dimmer, or not bright enough.
- c) metamers are color pairs that look or are perceived identically to the human eye, although they emit different light spectra, because they produce the same response in the three cone types (L, M, S)

2 Sheet 2

2.1 Assignment 1

- a)
 - Vector-based advantages:
 - resolution independent
 - Vector-based disadvantages:
 - image build-up is difficult in complex scenes
 - Raster-based advantages:
 - repeated image build-up independent of scene complexity
 - Raster-based disadvantages:
 - Moire effects due to finite number of pixels
- b) a spectral color, or a monochromatic color, is light produced by the light of single wavelength in the visible spectrum
- c)
- d) mach bands are an optical illusion where we perceive extra-bright and extra-dark bands near the boundary between two regions with different luminance gradients

a receptive field have usually a center-surround structure where light in the center excites a neuron, while the light in the surrounding region inhibits it, or vice versa.

Mach bands arise because these center-surround receptive fields perform a local contrast enhancement. Near a luminance edge, one receptive field may receive more excitation than inhibition, making the side look brighter, or vice versa. e) three operations: i)

relation to YIQ

in the YIQ model the Y channel is calculated approximately as:

$$Y = 0.3 R + 0.6G + 0.1 B$$

where we see that green contributes the most and blue the least. This corresponds roughly to the human perception of brightness - it is strongest around green-yellow region and weakest in the blue region, due to S cells contributing close to nothing to the perception of brightness.

I and Q channels carry color-difference information. They do not correspond exactly to the biological opponent channels, but they similarly separate color information from brightness detail - human eye is much more sensitive to contrast than to color difference

1) The light spectrum is given as

$$P(\lambda) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\lambda - \mu)^2}{2\sigma^2}\right) W$$

Is a Gaussian / normal distribution shape:

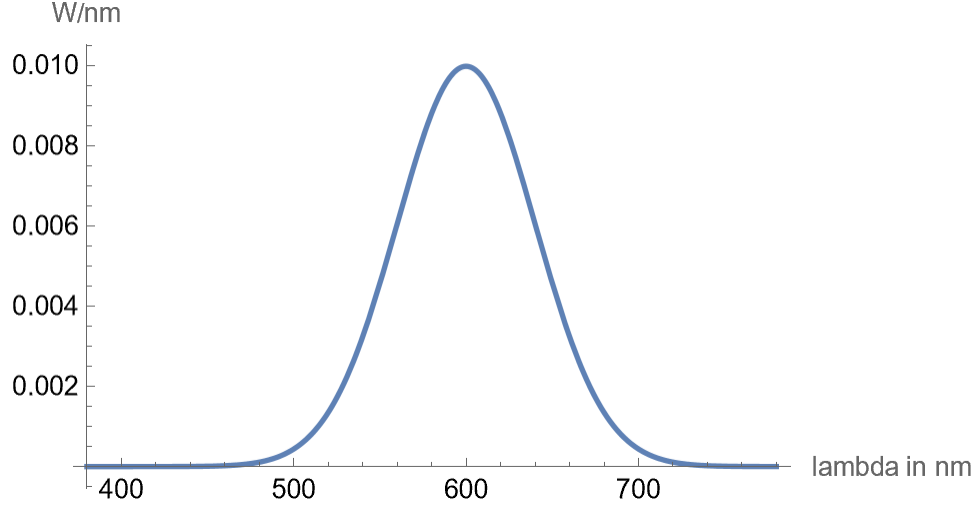


Figure 2.1: $P(\lambda)$

And the color matching functions are given as:

$$\bar{x}(\lambda) = 1.065 \exp\left(-\frac{1}{2} \left(\frac{\lambda - 595.8}{33.33}\right)^2\right) + 0.366 \exp\left(-\frac{1}{2} \left(\frac{\lambda - 446.8}{19.44}\right)^2\right)$$

$$\bar{y}(\lambda) = 1.014 \exp\left(-\frac{1}{2} \left(\frac{\ln \lambda - \ln 556.3}{0.075}\right)^2\right)$$

$$\bar{z}(\lambda) = 1.839 \exp\left(-\frac{1}{2} \left(\frac{\ln \lambda - \ln 449.8}{0.051}\right)^2\right)$$

which can be plotted as follows (together with a scaled version of $P(\lambda)$):

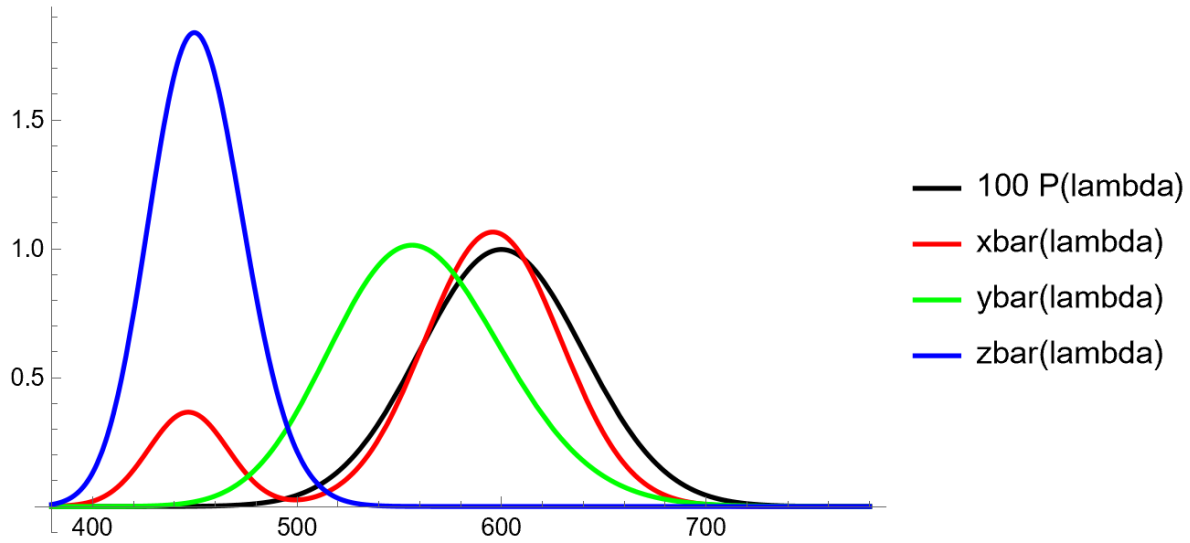


Figure 2.2: x, y, z

To compute X, Y, Z we compute the integrals:

$$X = \int_{\lambda} \bar{x}(\lambda) P(\lambda) d\lambda,$$

$$Y = \int_{\lambda} \bar{y}(\lambda) P(\lambda) d\lambda,$$

$$Z = \int_{\lambda} \bar{z}(\lambda) P(\lambda) d\lambda.$$

We can use a symbolic or numerical mathematics software. In our case we used mathematica where we computed the integrals with the following code:

First we define the functions:

```
sigma = 40;
mu = 600;

P[lambda_] := 1/Sqrt[2 Pi sigma^2] Exp[-(lambda - mu)^2/(2 sigma^2)];

xbar[lambda_] :=
  1.065 Exp[-1/2 ((lambda - 595.8)/33.33)^2] +
  0.366 Exp[-1/2 ((lambda - 446.8)/19.44)^2];

ybar[lambda_] :=
```

```

1.014 Exp[-1/2 ((Log[lambda] - Log[556.3])/0.075)^2];

zbar[lambda_] :=
1.839 Exp[-1/2 ((Log[lambda] - Log[449.8])/0.051)^2];

```

Then we compute the integrals with:

```

X = NIntegrate[xbar[lambda] P[lambda], {lambda, 380, 780}];
Y = NIntegrate[ybar[lambda] P[lambda], {lambda, 380, 780}];
Z = NIntegrate[zbar[lambda] P[lambda], {lambda, 380, 780}];

```

and find

$$\begin{aligned}
X &\approx 0.679966 \\
Y &\approx 0.569252 \\
Z &\approx 0.00561767
\end{aligned}$$

and finally we compute the x, y values as:

$$\begin{aligned}
x &= \frac{X}{X+Y+Z} \approx 0.541881 \\
y &= \frac{Y}{X+Y+Z} \approx 0.453642
\end{aligned}$$

To compute linear RGB from XYZ we use the following linear transformation:

$$\begin{pmatrix} R_{\text{lin}} \\ G_{\text{lin}} \\ B_{\text{lin}} \end{pmatrix} = \begin{pmatrix} 3.2406 & -1.5372 & -0.4986 \\ -0.9689 & 1.8758 & 0.0415 \\ 0.0557 & -0.2040 & 1.0570 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

We achieve this with the following mathematica code:

```

XYZtoLinearSRGB[{X_, Y_, Z_}] := {
  3.2406 X - 1.5372 Y - 0.4986 Z,
  -0.9689 X + 1.8758 Y + 0.0415 Z,
  0.0557 X - 0.2040 Y + 1.0570 Z
}

{R, G, B} = XYZtoLinearSRGB[{X, Y, Z}]

```

and obtain:

$$\begin{aligned}R &\approx 1.33 \\G &\approx 0.41 \\B &\approx -0.07\end{aligned}$$

Here R and B are outside of representable RGB gamut. Clipping them we get

$$\begin{aligned}R &\approx 1 \\G &\approx 0.41 \\B &\approx 0\end{aligned}$$

Converting this to $[0, 255]$ scale we get

$$(R, G, B) = (255, 171, 0)$$

This is a bright yellow-orange color, with no blue

3 ICG Mock Exam 2021 — Concise Solutions

3.1 Task 1 — Fundamentals

3.1.1 a) Raster graphics vs. vector graphics

Not yet completed.

3.1.2 b) Concepts

i) Up vector

The **up vector** specifies which direction appears upward in the camera image. Together with the view direction, it fixes the camera coordinate system and removes ambiguity about camera roll. It must not be parallel to the view direction.

ii) Double buffering

The **front buffer** displays the current image while the next frame is rendered into the invisible **back buffer**. After rendering, the buffers are swapped. This prevents incomplete frames and decouples rendering from screen refresh.

iii) View frustum

The **view frustum** is the region visible to the camera, bounded by left, right, top, bottom, near, and far planes. It is a truncated pyramid for perspective projection and a cuboid for orthographic projection. Geometry outside it is clipped; projection maps it to the normalized unit cube.

iv) Backface culling

Triangles facing away from the camera are discarded, usually based on vertex winding. For closed meshes these backfaces are hidden anyway, so culling saves rasterization work. It is unsuitable for visible interiors or two-sided surfaces.

3.1.3 c) Vertex normals and degenerate triangles

For each triangle, compute its face normal:

$$\mathbf{n}_f = (\mathbf{p}_2 - \mathbf{p}_1) \times (\mathbf{p}_3 - \mathbf{p}_1).$$

At each vertex, sum adjacent face normals—optionally area- or angle-weighted—and normalize. A zero-length edge yields a zero cross product, which cannot be normalized; degenerate triangles must therefore be ignored or removed.

3.1.4 d) Sharp edges with Gouraud and Phong shading

Multiple normals preserve a normal discontinuity across a sharp edge in both methods. **Gouraud shading** computes and interpolates vertex colors, so it may miss lighting effects between vertices. **Phong shading** interpolates the separate normals and evaluates illumination per fragment, giving more accurate results at higher cost.

3.1.5 e) Projection types

- **Orthographic projection:** parallel projection rays; size does not decrease with distance and parallel lines remain parallel.
- **Perspective projection:** rays meet at the projection centre; distant objects appear smaller.

3.1.6 f) Aliasing

Aliasing occurs when a signal is sampled below the Nyquist rate. High frequencies are then represented as false lower frequencies. Artifacts include jaggies, Moiré patterns, distorted detail, and temporal flickering.

3.1.7 g) Perspective division and dehomogenization

After projection, a point has homogeneous clip coordinates

$$(x_h, y_h, z_h, w_h).$$

Perspective division gives

$$(x_h, y_h, z_h, w_h) \mapsto \left(\frac{x_h}{w_h}, \frac{y_h}{w_h}, \frac{z_h}{w_h} \right).$$

Because w_h depends on depth in perspective projection, this produces foreshortening and maps the frustum to normalized device coordinates. It is precisely dehomogenization after projection, also called the normalization transformation.

3.1.8 h) Rendering equation

$$L(\mathbf{x}, \omega) = E(\mathbf{x}, \omega) + \int_{\Omega^+} f(\mathbf{x}, \omega_i \rightarrow \omega) L_{\text{in}}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i.$$

i) Components

- **A** — $L(\mathbf{x}, \omega)$: outgoing radiance from $\mathbf{x}$ in direction ω .
- **B** — $E(\mathbf{x}, \omega)$: emitted radiance.
- **C** — $f(\mathbf{x}, \omega_i \rightarrow \omega)$: fraction of incoming light reflected toward ω .
- **D** — $L_{\text{in}}(\mathbf{x}, \omega_i)$: incoming radiance from ω_i .

The integral sums all incoming directions in the positive hemisphere; $\cos \theta_i$ accounts for projected surface area.

ii) Name and dimensionality of f

f is the **bidirectional reflectance distribution function (BRDF)**. At a fixed point it is four-dimensional:

$$f(\theta_i, \phi_i, \theta_o, \phi_o).$$

iii) Isotropic BRDF

Rotational invariance around the normal removes one angular degree of freedom, leaving three dimensions:

$$f(\theta_i, \theta_o, \phi_i - \phi_o).$$

iv) Recursion

Incoming radiance at $\mathbf{x}$ is outgoing radiance from the next visible surface point. Evaluating it requires the same equation there, producing recursive secondary rays.

3.2 Task 2 — Color

3.2.1 a) Role of K in CMYK

K is **key/black ink**. It replaces the common black component

$$K = \min(C, M, Y),$$

because CMY inks do not produce a clean covering black. Separate black improves contrast and saves colored ink.

3.2.2 b) Missing dimension in the chromaticity diagram

The diagram omits **luminance** Y and displays only chromaticity (x, y) . Lowercase z is not an independent missing dimension because $z = 1 - x - y$.

i Recall: CIE XYZ and xyY

X, Y, Z are tristimulus values obtained from the spectrum $S(\lambda)$:

$$X = \int S(\lambda) \bar{x}(\lambda) d\lambda, \quad Y = \int S(\lambda) \bar{y}(\lambda) d\lambda, \quad Z = \int S(\lambda) \bar{z}(\lambda) d\lambda.$$

Y represents luminance. Chromaticity is

$$x = \frac{X}{X + Y + Z}, \quad y = \frac{Y}{X + Y + Z}, \quad z = \frac{Z}{X + Y + Z} = 1 - x - y.$$

3.2.3 c) Luminance sensitivity and compression

Vision is more sensitive to **luminance** than to chrominance variation. Compression therefore preserves luminance at higher resolution while subsampling or more strongly quantizing chrominance channels, e.g. C_b and C_r in YCbCr.

3.2.4 d) MacAdam ellipses

A MacAdam ellipse contains colors indistinguishable from its centre at a chosen threshold. Their varying sizes and orientations in CIE xy show perceptual non-uniformity. In a perfectly uniform color space they would be equal-sized circles.

3.2.5 e) Metamerism

Metamerism: different spectra produce the same cone responses and are therefore perceived as the same color.

3.2.6 f) Mach bands

Mach bands are illusory bright and dark bands near changes in a luminance gradient. Centre-surround receptive fields and lateral inhibition exaggerate local contrast: the bright side appears brighter and the dark side darker.

i Recall: centre-surround function

An ON-centre/OFF-surround field behaves approximately as

$$R \approx I_{\text{centre}} - \text{average}(I_{\text{surround}}).$$

Uniform illumination largely cancels, while local differences produce a strong response. This enhances edges and reduces redundant brightness information.

3.2.7 g) CIE chromaticity diagram

i) Gamut of n primaries

The gamut is the **convex hull** of the primaries: a convex polygon with at most n vertices; three primaries usually form a triangle.

ii) Spectral colors

Monochromatic colors lie on the curved **spectral locus**. The straight segment EF is excluded.

iii) Line of purples

The straight segment EF contains additive mixtures of the red and violet endpoints. These purples are not monochromatic; mixtures of two colors lie on the line connecting them.

iv) Point C and complementary colors

$C \approx (1/3, 1/3)$ is achromatic white. Since A, C, D are collinear and C lies between A and D , suitable proportions of A and D produce white:

$$C = (1 - t)A + tD.$$

Thus A and D are complementary relative to white point C .

v) Relation between A, B, and D

B lies on segment AD , so it is an additive mixture of A and D . Because it lies between A and white point C , it is a less saturated version of A .

3.3 Task 3 — Ray Tracing

3.3.1 a.i) Ray generation

For every pixel sample, generate a primary ray from camera position $\mathbf{e}$ through the corresponding image-plane point $\mathbf{s}$:

$$\mathbf{r}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e}), \quad t \geq 0.$$

The nearest intersection determines the visible surface and therefore the pixel color.